2026 Basic Abstract Algebra Study

PRE-COURSE SELF ASSESSMENT

SOLUTION - Q27

Version
v1.0.4

Instructor
Abiria

Published on
2026-05-11

Hosted by
Coding Lab

Read this carefully before moving on.

In the following questions, *prove or disprove* means you need to either show a correct logical proof of a given statement or provide at least one counterexample that disproves the statement. Describe each deduction step by step, citing the relevant definition or fact at each step.

As a tiny example of a proof answer, consider the following statement.

For any integers a and b, if a and b are both even, then so is $a + b$.

The above statement is true.

Proof. Assume a and b are even integers. By the definition of an even integer, there exist integers k and l such that $a = 2k$ and $b = 2l$. Then $a + b = 2k + 2l = 2(k + l)$, and since $k + l$ is an integer, $a + b$ is even by the same definition. □

As a tiny example of a disproof answer, consider the following statement.

For any integers a and b, if $a + b$ is even, then a and b are both even.

The above statement is false.

Proof. Take $a = 1$ and $b = 1$. Both are odd, but $a + b = 2$ is even. Hence the statement is false. □

It is highly encouraged to write down full sentences like the above samples, rather than just throwing out a few symbols or numbers without explaining why.

Note 1

The following symbols will be used throughout the following questions.

- $\mathbb{Z}$: A set of all integers.
- $\mathbb{Z}^+$: A set of all positive integers. i.e., $\mathbb{Z}^+ = \{1, 2, 3, \ldots\}$.
- $\mathbb{Z}_n$: A set of non-negative integers from 0 to $n - 1$ for some positive integer n. For instance, $\mathbb{Z}_4 = \{0, 1, 2, 3\}$.
- $\mathbb{Q}$: A set of all rational numbers.
- $\mathbb{R}$: A set of all real numbers.
- $\mathbb{R}^+$: A set of all positive real numbers.
- $\mathbb{C}$: A set of all complex numbers.

Definition 2 (Extensionality)

Two sets X and Y are said to be equal if and only if X is a subset of Y and Y is a subset of X. This is called the *axiom of extensionality*. If two sets X and Y are equal, we write $X = Y$.

Q1.

Note 3. *$A \setminus B$ denotes the set of all elements in A that are not in B.*

Which of the following is distinct from the others?

- ○ **1.** $\{-x \mid x \in \mathbb{R}\}$
- ○ **2.** $\{xx \mid x \in \mathbb{R}\}$
- ○ **3.** $\{xy \mid x, y \in \mathbb{R}\}$
- ○ **4.** $\{xxx \mid x \in \mathbb{R}\}$
- ○ **5.** $\left\{\frac{x}{y} \mid x \in \mathbb{R}, y \in \mathbb{R} \setminus \{0\}\right\}$

Q2.

Note 4. *The sum of two integers is always an integer.*

Show that $\{x + y \mid x, y \in \mathbb{Z}\} = \mathbb{Z}$ using extensionality. (Definition 2)

Hint 5. *First, show that $\{x + y \mid x, y \in \mathbb{Z}\} \subset \mathbb{Z}$. Then, show that $\mathbb{Z} \subset \{x + y \mid x, y \in \mathbb{Z}\}$. In the latter case, it is enough to show that for any integer, there exist at least a pair of integers such that their sum gets equal to that integer.*

> **Definition 6 (Size of finite sets)**
>
> A *size* of a finite set X is the number of elements it contains, and denoted as $|X|$.

Q3.

Let A be a set of all prime numbers that are even. What is the value of $|A|$?

- ○ **1.** 0
- ○ **2.** 1
- ○ **3.** 2
- ○ **4.** 3
- ○ **5.** 4

Q4.

Let $A = \{n + m \mid n, m \in \mathbb{Z}_{20261212}\}$. What is the value of $|A|$?

Hint 7. $|\{n + m \mid n, m \in \mathbb{Z}_2\}| = |\{0 + 0, 0 + 1, 1 + 0, 1 + 1\}| = |\{0, 1, 2\}| = 3$.

Answer: __________

Q5.

Let T be the set of all real numbers except 0 and 1. Define two functions $f, g : T \to T$ as $f(x) = x^{-1}$ and $g(x) = 1 - x$, respectively.

Let S be the set of all possible functions that can be created by composing f and g any number of times in any order, such as f, g, $g \circ f$, $f \circ g$, $g \circ g \circ f \circ f$, $f \circ g \circ f$, $f^{20261212}$, etc.

What is the value of $|S|$?

Answer: __________

Q6.

Let $\mathrm{Di}_6 = \{$⚀, ⚁, ⚂, ⚃, ⚄, ⚅$\}$ be a set of all possible faces of a six-sided die.

How many distinct ordered pairs can be made from Di_6?

○ **1.** 36 ○ **2.** 21 ○ **3.** 6 ○ **4.** 30 ○ **5.** 12

> **Definition 8 (Product of sets)**
>
> Let X, Y be two sets. A *product* $X \times Y$ between them is a set of all ordered pairs (x, y) where $x \in X$ and $y \in Y$.

Q7.

Let $X = \{10, 20, 30\}$ and $Y = \{\leftarrow, \rightarrow\}$. Find the same set as $Y \times X$.

○ **1.** $\{(10, 10), (20, 20), (30, 30), (\leftarrow, \leftarrow), (\rightarrow, \rightarrow)\}$
○ **2.** $\{(10, \leftarrow), (10, \rightarrow), (20, \leftarrow), (20, \rightarrow), (30, \leftarrow), (30, \rightarrow)\}$
○ **3.** $\{(\leftarrow, 20), (20, \leftarrow), (\leftarrow, 20), (20, \leftarrow), (30, \rightarrow), (\rightarrow, 30)\}$
○ **4.** $\{(10, \rightarrow), (20, \rightarrow), (30, \rightarrow), (10, \leftarrow), (20, \leftarrow), (30, \leftarrow)\}$
○ **5.** $\{(\leftarrow, 20), (\leftarrow, 30), (\leftarrow, 10), (\leftarrow, 20), (\rightarrow, 10), (\rightarrow, 20), (\rightarrow, 30)\}$

> **Definition 9 (Power set)**
>
> A set of all subsets of a set X is called a *power set* of X, and denoted as $\mathcal{P}(X)$.

Q8.

Which of the following is the same as $\mathcal{P}(\{a, 1, *\})$, if a, 1, and $*$ are all distinct?

○ **1.** $\{\{\varnothing, \{a\}, \{1\}, \{*\}, \{a, 1\}, \{1, *\}, \{a, *\}, \{a, 1, *\}\}\}$
○ **2.** $\{\{*, 1, a\}, \{*, a\}, \{*, 1\}, \{1, a\}, \{*\}, \{a\}, \{1\}\}$
○ **3.** $\{\varnothing, \{1\}, \{a\}, \{*\}, \{(a, 1)\}, \{(1, *)\}, \{(a, *)\}, \{(1, a, *)\}\}$
○ **4.** $\{\{*, 1, a\}, \{a\}, \{a, 1, a\}, \{*, *\}, \{1\}, \varnothing, \{1, *\}, \{a\}, \{a, *, *\}\}$
○ **5.** $\{\varnothing, \{a\}, \{1\}, \{*\}, \{a, 1\}, \{1, a\}, \{1, *\}, \{*, 1\}, \{a, 1, *\}\}$

Q9.

What is the value of $|\mathcal{P}(\mathbb{Z}_2 \times \mathbb{Z}_5)|$?

Answer: ___________

Q10.

Prove or disprove that the following holds for any two sets A and B.

$$(A \setminus B) \cap (B \setminus A) = \varnothing$$

Q11.

Prove or disprove that the following holds for any set X.

$$\mathcal{P}(X) \cap X = \varnothing$$

> **Definition 10 (Singleton)**
>
> A set is called a *singleton* if it has only one element.

Q12.

Prove or disprove the following statement.

A set is a singleton if and only if it has only one proper subset.

> **Definition 11 (Injection, Surjection, and Bijection)**
>
> A function $f : X \to Y$ is said to be *injective* or called an *injection* if for every $x, y \in X$, if $x \neq y$ then $f(x) \neq f(y)$. A function is said to be *surjective* or called a *surjection* if for every $y \in Y$, there exists an $x \in X$ such that $f(x) = y$. A function is said to be *bijective* or called a *bijection* if it is both an injection and a surjection.

Q13.

For a set X, think of another set $X/{=}$ that is a set of all singletons from all elements of X. For instance, if X is $\{a, b, c\}$, then $X/{=}$ becomes $\{\{a\}, \{b\}, \{c\}\}$.

Prove or disprove that there is a bijection between X and $X/{=}$.

Definition 12 (Equality of functions)

Two functions with the same domain and same codomain are said to be *equal* if and only if they map every element of the domain to the same element of the codomain.

In other words, if $f, g : X \to Y$ are functions with the same domain and codomain, then $f = g$ if and only if $f(x) = g(x)$ for every $x \in X$.

Q14.

Show that for any singleton, there is always a unique function from any set to it.

Hint 13. *Let $\{*\}$ be a singleton and X be any set, and consider a function $f : X \to \{*\}$. First show that such function always exists, and then prove that if there is a function $f' : X \to \{*\}$, then $f = f'$.*

Q15.

Note 14. *An induction here refers to the mathematical induction.*

Let A_i be a singleton for $i = 1, ..., n$ for a positive integer n. Show by induction that $A_1 \times ... \times A_n$ is also a singleton.

Q16.

Let X, Y be finite sets. How many functions $f : X \to Y$ are there from X to Y?

- ○ **1.** $|Y|^{|X|}$
- ○ **2.** $2^{|X| \cdot |Y|}$
- ○ **3.** ${}_{|Y|}P_{|X|}$
- ○ **4.** $|X|^{|Y|}$
- ○ **5.** $|X|! \cdot |Y|!$

Q17.

Prove or disprove that the following statement holds for any finite sets X, Y and Z.

If the number of bijections from X to Y is the same as the number of bijections from X to Z, then $|Y| = |Z|$.

Q18.

Prove or disprove that if f and g are injective, then $g \circ f$ is injective for any functions f and g.

Q19.

Prove or disprove that if $f \circ g = f \circ h$, then $g = h$ for any functions $f : Y \to Z$, $g, h : X \to Y$.

Q20.

Prove or disprove the following statement.

For any finite set X and Y where $|X| = |Y|$, a function $f : X \to Y$ is injective if and only if it is surjective.

Hint 15. *Use the pigeonhole principle.*

Q21.

Note 16. $\ln(x)$ *denotes the natural logarithm of* x*.*

Prove or disprove that $\ln : \mathbb{R}^+ \to \mathbb{R}$ is bijective.

Q22.

Which of the following is the same as $(-i)^{4443}$?

○ **1.** $(-i)^{666}$ ○ **2.** i^{321} ○ **3.** i^{123} ○ **4.** $(-i)^{4812}$ ○ **5.** $(-i)^{333}$

Q23.

How many trailing zeros are there in the decimal representation of $629!$?

Hint 17. *There are two trailing zeros in* $11!$*, as* $11! = 39916800$.

Answer: __________

Q24.

Note 18. *For integers* a *and* b*,* $a \perp b$ *denotes that there are no common divisors of* a *and* b *other than 1. It is equivalent to saying that* $\gcd(a, b) = 1$*. In contrast,* $a \not\perp b$ *denotes that there are common divisors other than 1 between* a *and* b*.*

(a) Prove or disprove that the following statement holds for any integers a, b and c.

If $a \perp b$ and $b \perp c$, then $a \perp c$.

(b) Prove or disprove that the following statement holds for any integers a, b and c.

$$\text{If } a \not\perp b \text{ and } b \not\perp c \text{, then } a \not\perp c.$$

Q25.

(a) Prove or disprove that the following statement holds for any integers a, b and c.

$$\text{If } a \text{ divides } b \text{ and } b \text{ divides } c\text{, then } a \text{ divides } c.$$

(b) Prove or disprove that the following statement holds for any integers a, b and c.

$$\text{If } a \text{ does not divide } b \text{ and } b \text{ does not divide } c\text{, then } a \text{ does not divide } c.$$

Lemma 19 (Euclid's lemma)

For integers a, b, c, if $a \perp b$ and a divides bc, then a divides c.

Q26.

Note 20. *For a rational number r, an irreducible form of r is a representation $r = \frac{p}{q}$ such that $p \in \mathbb{Z}$, $q \in \mathbb{Z}^+$, and $p \perp q$.*

Define $f : \mathbb{Q} \to \mathbb{Z} \times \mathbb{Z}^+$ to be the rule that sends each $r \in \mathbb{Q}$ to (p, q), where $\frac{p}{q}$ is an irreducible form of r.

(a) Show that f is a well-defined function.

Hint 21. *First, show that every rational number admits an irreducible form. Then, show that if $\frac{p}{q}$ and $\frac{p'}{q'}$ are both irreducible forms of the same rational number, then $p = p'$ and $q = q'$.*

(b) Prove or disprove that f is injective.

(c) Prove or disprove that f is surjective.

Q27.

Show by induction that an equation of the following form has a unique complex solution for each $n \in \mathbb{Z}^+$.

$$x^n = 0$$

Hint 22. *If $ab = 0$, then either a or b is zero.*

The solution of such an equation is always 0, and we show it by induction.

Proof. If $n = 1$, then the solution of the equation $x^1 = x = 0$ is trivially 0, and it is the only solution. It can be seen that 0 is indeed the solution of the equation, as $0^1 = 0$.

Assume that the equation $x^n = 0$ has a unique solution, namely 0. If an equation $x^{n+1} = 0$ is given, it follows that either $x = 0$ or $x^n = 0$ as $x^{n+1} = x^n x = 0$. If $x = 0$, then it shows that the solution is 0. If $x^n = 0$, then by the induction hypothesis, the unique solution of it is 0, which also shows that the solution of $x^{n+1} = 0$ is 0. As the possible solution is 0 in either case, the solution is 0 and unique. It can easily be verified that 0 is indeed a solution, as $0^{n+1} = 0$. □

We have used the property that if $ab = 0$, then either a or b is zero. Such a property is a characteristic property of integral domains. As we already know that it holds in complex numbers, we can deduce that they form an integral domain.

Q28.

Let $\boldsymbol{X}$ be a $m \times m$ matrix with complex entries and $\mathbf{0}$ be the zero matrix. Prove or disprove that an equation of the following form has a unique solution for each $n, m \in \mathbb{Z}^+$.

$$\boldsymbol{X}^n = \mathbf{0}$$

Q29.

Let $\boldsymbol{A}$ be a $n \times n$ matrix with complex entries. Prove or disprove that the following equation has a unique solution for each $n \in \mathbb{Z}^+$.

$$\boldsymbol{A} = -\boldsymbol{A}$$

Q30.

Prove or disprove that for any positive integer $n \in \mathbb{Z}^+$, an equation of the following form has at most two distinct solutions in $\mathbb{Z}_n$.

$$x^2 \equiv 1 \pmod{n}$$

Hint 23. *By 'at most two distinct' solutions, we mean that the number of solutions can be zero, one, or two, but not more than two.*

Q31.

Given any $a \in \mathbb{Z}_n$ such that $a \perp n$, prove or disprove that an equation of the following form has a unique solution in $\mathbb{Z}_n$ for each $n \in \mathbb{Z}^+$.

$$ax \equiv 1 \pmod{n}$$

Hint 24. *See **Q20**. Can you find a relevant bijection on $\mathbb{Z}_n$?*

Q32.

(a) How many solutions does the equation below have in $\mathbb{Z}_{12121212}$?

$$1212x \equiv 4848 \pmod{12121212}$$

Hint 25. *Try considering similar questions on $\mathbb{Z}_3$, $\mathbb{Z}_8$, or any small $\mathbb{Z}_n$ and solving them. Can you find a pattern?*

Answer:__________

(b) How many solutions does the equation below have in $\mathbb{Z}_{20261212}$?

$$1212x \equiv 2026 \pmod{20261212}$$

Answer:__________

(c) How many solutions does the equation below have in $\mathbb{Z}_{20262121}$?

$$1212x \equiv 2026 \pmod{20262121}$$

Hint 26. *See **Q31**.*

Answer:__________

Q33.

Let $\mathbb{Z}^{\mathbb{Z}}$ be a set of all functions from $\mathbb{Z}$ to $\mathbb{Z}$ and define a function $f \in \mathbb{Z}^{\mathbb{Z}}$ by $f(x) = 2x$. We denote the identity function on $\mathbb{Z}$ by $1_{\mathbb{Z}}$.

Prove or disprove that the following equation has a unique solution in $\mathbb{Z}^{\mathbb{Z}}$. In other words, there is a unique function $x : \mathbb{Z} \to \mathbb{Z}$ such that the following equation holds.

$$x \circ f = 1_{\mathbb{Z}}$$

Q34.

Let Γ be an arbitrary finite set of invertible $n \times n$ matrices with complex entries with $n \in \mathbb{Z}^+$. Suppose that Γ is closed under matrix multiplication. In other words, for any $\boldsymbol{A}, \boldsymbol{B} \in \Gamma$, the product $\boldsymbol{AB}$ is also in Γ.

Let $\mathbf{\Psi}$ be the sum of all matrices in Γ.

$$\mathbf{\Psi} = \sum_{\mathbf{\Omega} \in \Gamma} \mathbf{\Omega}$$

Show that for any matrix $\mathbf{\Lambda} \in \Gamma$, the equation $\mathbf{\Lambda}\mathbf{\Psi} = \mathbf{\Psi}$ holds.

Hint 27. *See **Q20**.*

End.

You may now submit your work.

Evaluation sheet

For each question so far, evaluate it based on the following criteria:

- If you encountered new concepts or new words for the first time, mark ✓ in the New concepts column.
- If you failed to understand the question, mark × in the Evaluation column.
- If you understood the question, but didn't know how to solve it, mark ? in the Evaluation column.
- If you understood the question and are sure you have solved it, mark ○ in the Evaluation column.

No.	New concepts	Evaluation	No.	New concepts	Evaluation
Q1			Q18		
Q2			Q19		
Q3			Q20		
Q4			Q21		
Q5			Q22		
Q6			Q23		
Q7			Q24		
Q8			Q25		
Q9			Q26		
Q10			Q27		
Q11			Q28		
Q12			Q29		
Q13			Q30		
Q14			Q31		
Q15			Q32		
Q16			Q33		
Q17			Q34		

Please answer these additional questions:

(a) Which was the hardest for you, among those you have solved? ___________

(b) Which seemed to be the hardest for you, among those you haven't solved? ___________

(c) Which was your favorite question, if you had to choose one? ___________